The next five chapters use D3 in the ways we typically think of. Chapter 4 teaches you how to create simple graphics from data, such as line charts, axes, and boxplots. Chapter 5 gives an in-depth exploration of various traditional data visualization layouts like pie charts, tree layouts, and word clouds. Chapter 6 is devoted to network visualization, which might seem exotic, but network visualization is being used more and more in a variety of domains. Chapter 7 dives into the rich mapping capabilities in D3, and includes leveraging TopoJSON to do interesting geodata manipulation in the browser. Chapter 8 is devoted to manipulating traditional HTML elements, like paragraphs and lists, to demonstrate that D3 is not tied to SVG.

The last three chapters and chapter 12 (online only) cover topics that can be considered deep diving into D3. I've found that each has become an important part of my own practice. This includes principles for wiring up your own data dashboard, creating your own D3 layouts and components, optimizing data visualization for large datasets, and writing data visualization for mobile. Even if you don't think you'll ever be using D3 in these ways, each of these chapters still touches on key aspects of using D3.

How to use this book

If you're just getting started with D3, I suggest going through chapters 1 through 4 in order. Each chapter builds on the last and establishes the basic principles not only of D3 but also of data visualization. After that, it depends on what you plan to use D3 for. If your data is mostly geographic, then you can jump to chapter 7, and similarly, if your data is mostly network data, you can jump to chapter 6. If you're doing traditional data visualization, then I suggest going to chapter 5 and then on to chapter 9 to start thinking about dashboards, which are a key component of traditional data visualization.

If you've been using D3 for a while and want to improve your skills, I suggest skimming the first three chapters. The parts that I think might be of particular interest are in chapter 3, and deal with color and loading external resources like SVG icons or HTML content. You might also want to review generators and components in chapter 4 to fill in any gaps you might have dealing with these common, but often underexamined, parts of D3. After that, it depends on what you see as your strengths and what you see as your goals for using D3. If you want to maximize traditional data visualization, take a look at chapter 5 to see the layouts, and then look at chapter 9 for dashboards. You're probably familiar with most of the content there, but these chapters deal with it more exhaustively than you likely have experienced. After that, look at chapter 11 and see if there are any optimization techniques you might want to bring into your data visualization, or look at chapter 8 and think about how you might use the D3 tricks you know to build UI elements and otherwise do traditional web development.

Much of the value of this book comes in chapters 6 and 7, which go into great detail about using D3 for two major areas of data visualization: networks and maps. Along those lines, the use of HTML5 canvas in chapters 8 and 11 is an area that even experienced D3 developers might not be familiar with.